VF Asia Footwear Laboratory Testing Manual

Test Method for Moisture Content	Test method # VF- E12 Page 1 of 1
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PURPOSE AND SCOPE: To measure the quantity of moisture contained in a test material.

REFERENCE: ASTM D3790

EQUIPMENT AND MATERIAL:

1. A mechanical convection oven capable of maintaining a temperature of +/-2°C
2. A thermometer with an accuracy of +/-0.5 ^{o}C for monitoring the oven temperature.
3. A balance capable of weighing up to 100 grams with an accuracy of +/-0.001grams.
4. Weighing dish which is large enough to hold the test sample.
5. Desiccator containing any normal desiccating agent, such as calcium sulfate, calcium chloride, or silica g
6. Leather, fabric, foam, plastic, rubber, and paper/fiber board.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

2. Die or hand cut a sample of the test material which has a weight of 5+/-1 gram

REPETITIONS: Test two (2) samples per material**.**

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

1. Determine the weight of the weighing dish to the nearest 0.001 gram**(W1)**
2. Place the test sample in the weighing dish ,and determine the weight of the test sample and weighing dish to the nearest 0.001 gram **(W2)**
3. Place the weighing dish and test sample in an oven at 100+/-2 ^{o}C for 16.0+/-0.5 hours.

Note: This can most easily be accomplished by placing the sample in the oven at the very end of the day, and removing at the start of the next day.

4. Place the weighing dish and test sample in the desiccator and allowed to cool for a minimum of 2 hours.
5. Immediately upon removal from the desiccator, determine the weight of the weighing dish and test sample to the nearest 0.001 grams **(W3)**

CALCULATION:

$$\text{Moisture content (\%)} = \frac{W2-W3}{W2-W1} * 100$$

REPORT: The average of the two moisture content measurements to the nearest 0.1%.